

# Fall 2006

**Problem 1 (Fa06, Sp08).** Compute  $\lim_{x \rightarrow 0} \frac{d^4}{dx^4} \frac{x}{\sin x}$ .

**Problem 2 (Fa06).** Let  $A = \begin{pmatrix} 3 & -2 \\ 1 & 0 \end{pmatrix}$ . Compute  $e^A = \sum_{n=0}^{\infty} \frac{A^n}{n!}$ .

**Problem 3 (Fa06).** Let  $U$  be a connected open subset of  $\mathbb{C}$  containing  $-2$  and  $0$ . Suppose that  $f: U \rightarrow \mathbb{C}$  is a holomorphic function whose Maclaurin expansion is

$$\sum_{n \geq 0} \binom{2n}{n} z^n$$

Prove that  $f(-2) \in \{1/3, -1/3\}$ .

**Problem 4 (Fa06).** Let  $R$  be a finite commutative ring without zero divisors and containing at least one element other than  $0$ . Prove that  $R$  is a field.

**Problem 5 (Fa06).** Let  $\mathcal{C}_{[0,1]}^0$  be the vector space over  $\mathbb{R}$  consisting of continuous functions from  $[0, 1]$  to  $\mathbb{R}$ . Show that the linear operator  $T: \mathcal{C}_{[0,1]}^0 \rightarrow \mathcal{C}_{[0,1]}^0$  defined by

$$(Tf)(x) = \int_0^x f(y) dy$$

has no nonzero eigenvectors.

**Problem 6 (Fa06).** Let  $p$  be prime. Prove that the polynomial  $f(x) = x^p - x + 1$  is irreducible over the field  $\mathbb{F}_p$  with  $p$  elements.

**Problem 7 (Fa06).** Prove that for every  $a \in \mathbb{C}$  and integer  $n \geq 2$ , the equation  $1 + z + az^n = 0$  has at least one root in the disc  $|z| \leq 2$ .

**Problem 8 (Fa06).** Let  $\mathbb{Z}$  denote the ring of integers and consider the linear map  $\mathbb{Z}^3 \rightarrow \mathbb{Z}^3$  defined by the  $3 \times 3$  matrix

$$A = \begin{pmatrix} 6 & 9 & 12 \\ 6 & 9 & 12 \\ 12 & 18 & 24 \end{pmatrix}$$

Compute the structure of the three abelian groups  $\ker(A)$ ,  $\text{img}(A)$ , and  $\text{coker}(A) = \mathbb{Z}^3 / \text{img}(A)$ . In particular, in each case determine whether the group is free abelian. If yes, give a basis.

**Problem 9 (Fa02, Fa06).** Let  $\mathbb{F}$  be a field such that the additive group of  $\mathbb{F}$  is finitely generated as a group. Prove that  $\mathbb{F}$  is finite.

**Problem 10 (Fa06).** Let  $f : \mathbb{C} \rightarrow \mathbb{C}$  be an entire function. Assume that  $|f(z^2)| \leq 2|f(z)|$  for all  $z \in \mathbb{C}$ . Show that  $f$  is constant.

**Problem 11 (Fa06).** Let  $\mathcal{C}_{[0,1]}$  be the vector space over  $\mathbb{R}$  consisting of continuous functions from  $[0, 1]$  to  $\mathbb{R}$ . Show that the functions  $1, x, x^2, \dots$  are linearly independent in  $\mathcal{C}_{[0,1]}$ .

**Problem 12 (Sp89, Fa06, Sp23).** Let  $f : \mathbb{R} \times [0, 1] \rightarrow \mathbb{R}$  be a continuous function. For  $x \in \mathbb{R}$ , define

$$g(x) = \max\{f(x, y) \mid y \in [0, 1]\}$$

Show that  $g$  is continuous.

**Problem 13 (Fa06).** Let  $f(x) \in \mathbb{Q}[x]$  be an irreducible polynomial. Suppose there is a field extension  $\mathbb{F}$  of  $\mathbb{Q}$  containing a zero  $a$  of  $f(x)$  such that  $\mathbb{F}$  does not contain any cubic root of  $a$ . Show that  $f(x^3)$  is irreducible over  $\mathbb{Q}$ .

**Problem 14 (Fa06).** Let  $f$  and  $g$  be entire functions such that

$$\int_{|z|=1} \frac{f(z)}{\sin^m z} dz = \int_{|z|=1} \frac{g(z)}{\sin^m z} dz$$

for all positive integers  $m$ . Prove that  $f = g$ .

**Problem 15 (Fa06).** Let  $G$  be a non-abelian group of order 21. Find the largest positive integer  $n$  with the property that whenever  $G$  acts on a set  $S$  of size  $n$ , some element of  $S$  is fixed by every element of  $G$ .

**Problem 16 (Fa06).** Let  $X$  and  $Y$  be metric spaces, and let  $f_1, f_2, \dots$  be continuous functions from  $X$  to  $Y$ . Suppose that the sequence  $\{f_n\}$  converges uniformly to a function  $f$ . Show that  $f$  is continuous.

**Problem 17 (Fa06).** Let  $A$  be an  $n \times n$  Hermitian matrix and  $B$  an  $n \times n$  positive definite matrix. Prove that there is an invertible complex  $n \times n$  matrix  $S$  such that  $S^*AS$  is diagonal and  $S^*BS = I$ .

**Problem 18 (Fa06).** Let  $z_0, z_1, \dots$  be a sequence of complex numbers such that  $z_{n+1} = 1 + 1/z_n$  for all  $n \geq 0$ . Prove that the sequence is convergent.